TASK 2 - KEY INSIGHTS SUMMARY

1. OVERALL CHURN RATE:

- 26.5% of customers churned = 1,869 out of 7,043 customers

2. GENDER:

- Gender has low impact on churn
- Males and females churn are at similar rates

3. TENURE:

- Churned customers stay at an average of 18 months
- Loyal customers stay at an average of 37.6 months
- New customers are the highest risk group

4. CONTRACT TYPE:

- Month-to-month =42.7% churn
- One year = 11.3% churn
- Two year = 2.8% churn
- Longer contracts = much lower churn

5. PAYMENT METHOD:

- Electronic check = 45.3% churn
- Credit card = 15.2% churn
- Bank transfer = 16.7% churn
- Auto-payment customers are more loyal.

In [2]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('Cleaned_Churn_Data.csv')

print("Data loaded")
print("Shape:", df.shape)
df.head()
```

```
Data loaded
Shape: (7043, 25)
```

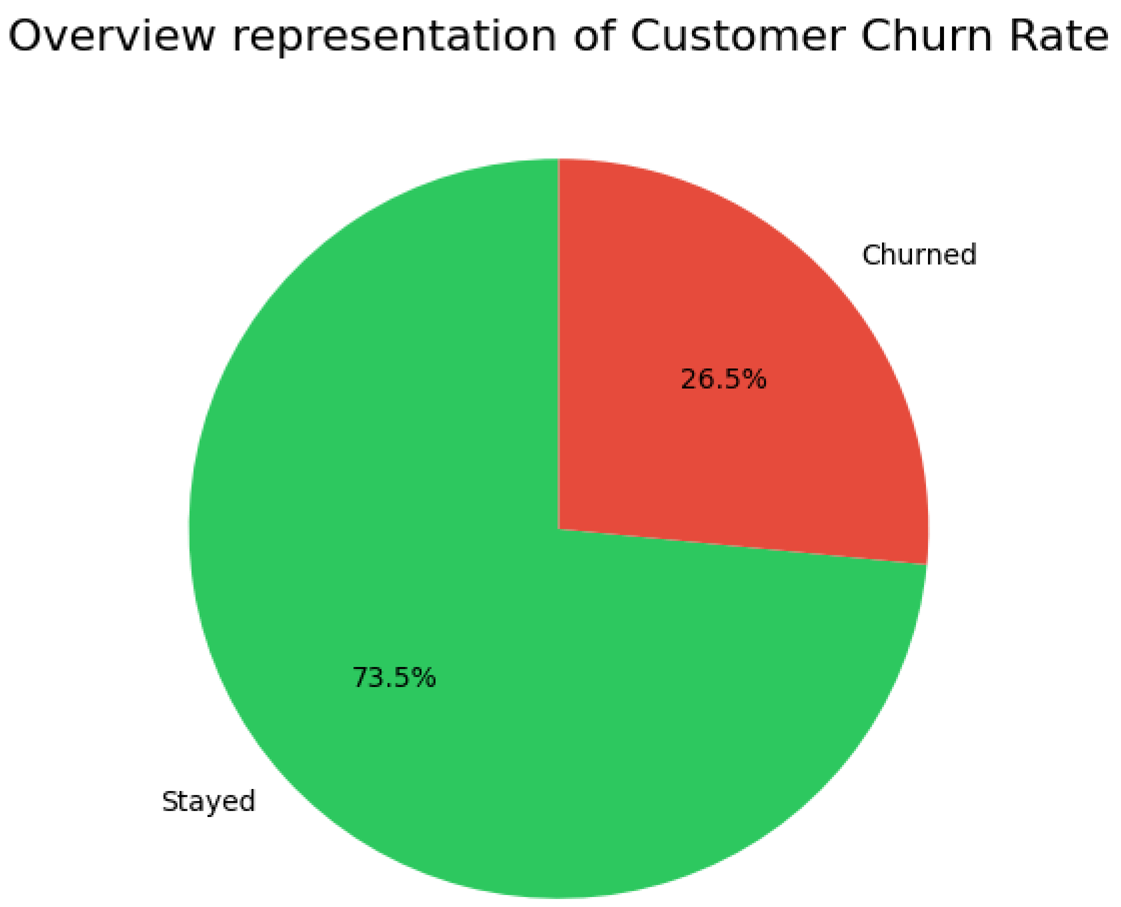
Out[2]:

	customerID	SeniorCitizen	tenure	MultipleLines	OnlineSecurity	OnlineBackup	Dev
0	7590-VHVEG	0	1	No phone service	No	Yes	
1	5575-GNVDE	0	34	No	Yes	No	
2	3668-QPYBK	0	2	No	Yes	Yes	
3	7795-CFOCW	0	45	No phone service	Yes	No	
4	9237-HQITU	0	2	No	No	No	

5 rows × 25 columns

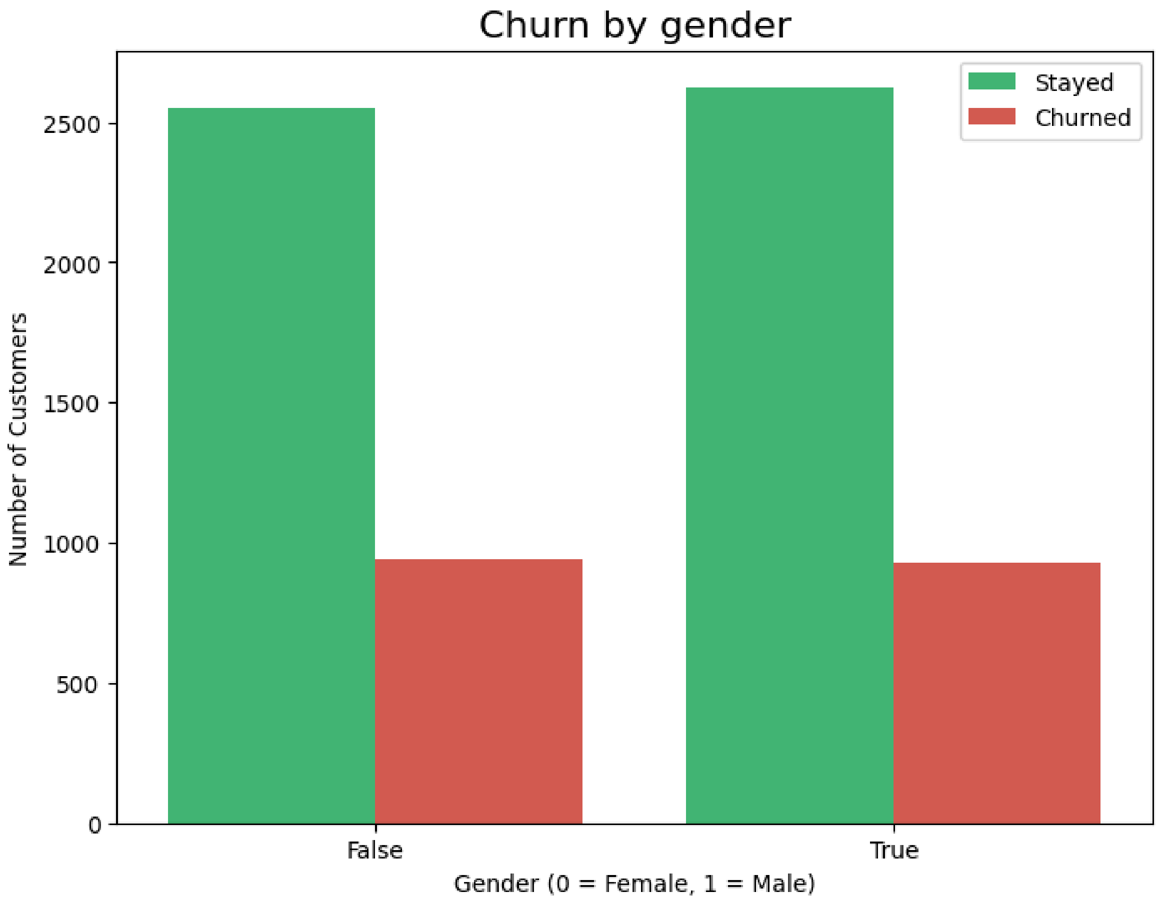
In [4]:

```
Churn_counts = df['Churn'].value_counts()
Churn_labels = ['Stayed', 'Churned']
Churn_colors = ['#2ecc61', '#e74c3c']

plt.figure(figsize=(8, 6))
plt.pie(Churn_counts, labels = Churn_labels, colors = Churn_colors, autopct = '%
plt.title('Overview representation of Customer Churn Rate', fontsize = 16)
plt.show()

total = len(df)
churned = df['Churn'].sum()
print(f"Total customers = {total}")
print(f"Churned = {churned} ({churned/total*100: .1f}%)")
print(f"Stayed  = {total-churned} ({(total-churned)/total*100:.1f}%)")
```




```
Total customers = 7043
Churned = 1869 ( 26.5%)
Stayed  = 5174 (73.5%)
```

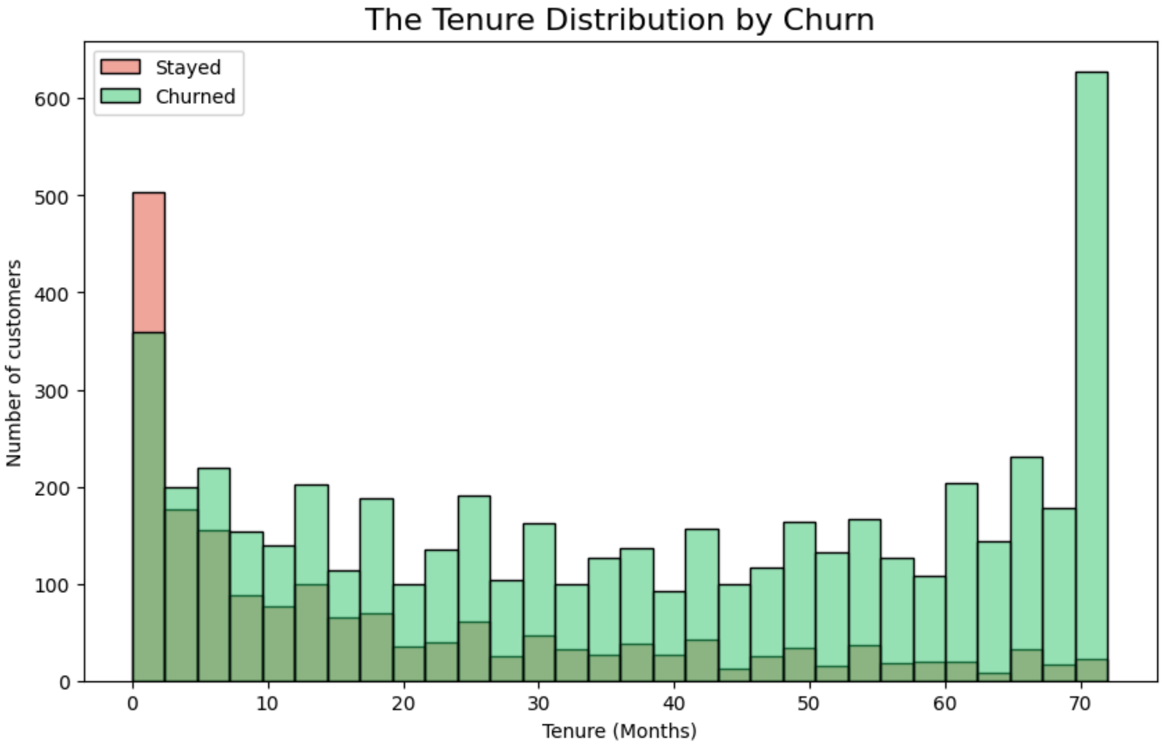
In [8]:
```
plt.figure(figsize = (8, 6))
sns.countplot(data = df, x ='gender_Male', hue = 'Churn', palette = ['#2ecc71',
plt.title('Churn by gender', fontsize = 16)
plt.xlabel('Gender (0 = Female, 1 = Male)')
plt.ylabel('Number of Customers')
plt.legend(labels=['Stayed', 'Churned'])
plt.show()

print("Churn by Gender:")
print(df.groupby('gender_Male')['Churn'].value_counts())
```




```
Churn by Gender:
gender_Male  Churn
False        0        2549
             1         939
True         0        2625
             1         930
Name: count, dtype: int64
```

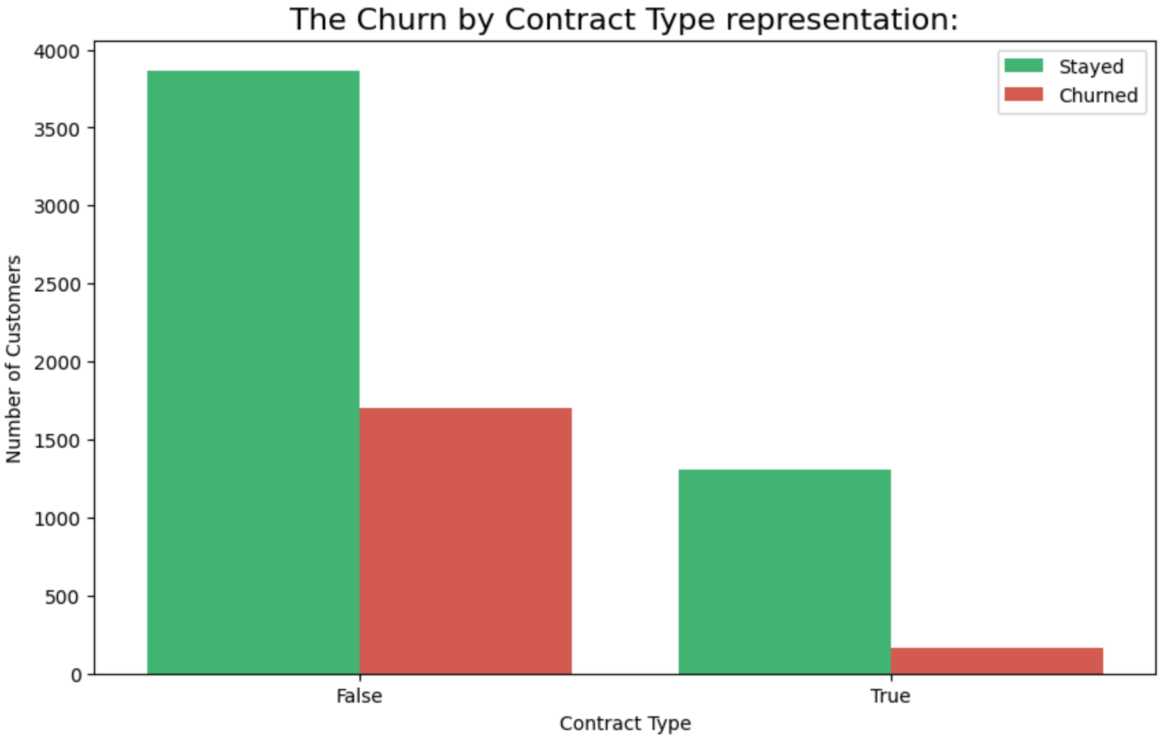
In [11]:

```
plt.figure(figsize = (10, 6))
sns.histplot(data = df, x = 'tenure', hue = 'Churn', bins = 30, palette = ['#2ec
plt.title('The Tenure Distribution by Churn', fontsize = 16)
plt.xlabel('Tenure (Months)')
plt.ylabel('Number of customers')
plt.legend(labels = ['Stayed', 'Churned'])
plt.show()

print("Average tenure for churned customers:", round(df[df['Churn'] == 1]['tenur
print("Average tenure for stayed customers:", round(df[df['Churn'] == 0]['tenure
```



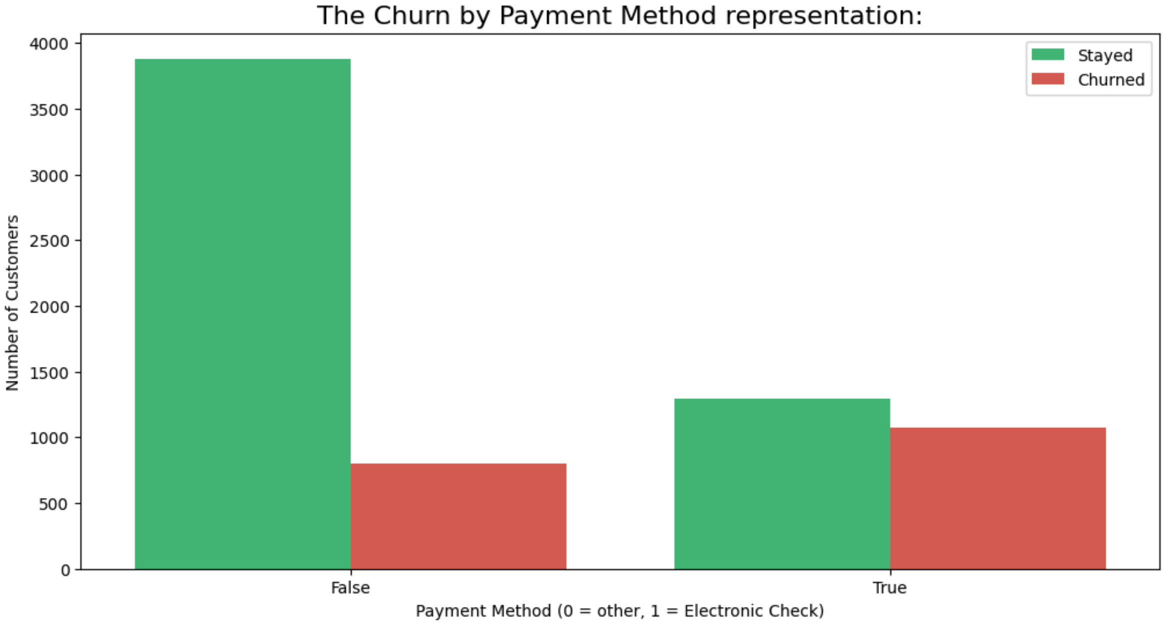

```
Average tenure for churned customers: 18.0 months
Average tenure for stayed customers: 37.6 months
```

In [13]:
```
plt.figure(figsize = (10, 6))
sns.countplot(data = df, x = 'Contract_One year', hue = 'Churn', palette = ['#2e

plt.title('The Churn by Contract Type representation:', fontsize = 16)
plt.xlabel('Contract Type')
plt.ylabel('Number of Customers')
plt.legend(labels = ['Stayed', 'Churned'])
plt.show()

print("Churn rate by Contract Type:")
contract_churn = df.groupby(['Contract_One year', 'Contract_Two year'])['Churn']
print(contract_churn.round(1))
```




```
Churn rate by Contract Type:
Contract_One year  Contract_Two year
False              False                42.7
                   True                  2.8
True               False                11.3
Name: Churn, dtype: float64
```

In [17]:
```
plt.figure(figsize = (12, 6))
sns.countplot(data = df, x = 'PaymentMethod_Electronic check', hue = 'Churn', pa

plt.title('The Churn by Payment Method representation:', fontsize = 16)
plt.xlabel('Payment Method (0 = other, 1 = Electronic Check)')
plt.ylabel('Number of Customers')
plt.legend(labels = ['Stayed', 'Churned'])
plt.show()

print("Churn rate by Payment Method:")
payment_cols = ['PaymentMethod_Credit card (automatic)', 'PaymentMethod_Electron
for col in payment_cols:
    method_name = col.replace('PaymentMethod_', '')
    churn_rate = df[df[col] == 1]['Churn'].mean()*100
    print(f"{method_name}: {churn_rate:.1f}%")

bank_mask = ((df['PaymentMethod_Credit card (automatic)'] == 0) &
             (df['PaymentMethod_Electronic check'] == 0) &
             (df['PaymentMethod_Mailed check'] == 0))
print(f"Bank transfer (automatic): {d[bank_mask]['Churn'].mean()*100:.1f}%")
```




```
Churn rate by Payment Method:
Credit card (automatic): 15.2%
Electronic check: 45.3%
Mailed check: 19.1%
Bank transfer (automatic): 16.7%
```